

Table 1: Paired entropy-regularisation comparisons across nine datasets, two network views, and two regulariser arms. Δpp is mean (treatment – baseline) validation-accuracy shift in percentage points; $W/L/T$ is wins/losses/ties across seeds; σ_b, σ_t are val-acc standard deviations (pp) of each arm; ΔH is the final spectral-entropy shift. Significance marks: *** $p<0.001$, ** $p<0.01$, * $p<0.05$, $\cdot$ $0.05\leq p<0.10$. All runs share the anchor MNIST config (Adam, $lr = 10^{-3}$) with $\lambda=0.1$ (SCALAR_ENTROPY) or $\lambda=10$ (KL_TRAJECTORY).

Dataset	View	Arm	n	ep	Δpp	t	W	L	T	$\sigma_b \rightarrow \sigma_t$	ΔH
MNIST (plain MLP)	dataflow	SCALAR_H	15	30	+0.165*	+2.45	10	5	0	0.285→0.163	−0.044
	factor	SCALAR_H	15	30	+0.157 $\cdot$	+1.84	10	5	0	0.285→0.177	−0.273
MNIST (ResMLP-20)	dataflow	KL_TRAJ	66	5	+0.049 $\cdot$	+1.98	36	28	2	0.329→0.310	−0.000
	dataflow	SCALAR_H	66	5	+0.046 $\cdot$	+1.76	36	29	1	0.329→0.286	−0.005
	dataflow	SCALAR_H	33	5	+0.070 $\cdot$	+1.89	18	15	0	0.328→0.294	−0.006
FashionMNIST	dataflow	SCALAR_H	33	15	+0.039	+1.13	19	13	1	0.347→0.307	−0.014
	factor	SCALAR_H	33	15	−0.009	−0.24	17	16	0	0.347→0.333	−0.038
KMNIST	dataflow	KL_TRAJ	33	5	−0.029	−1.00	15	17	1	0.898→0.844	−0.000
	dataflow	SCALAR_H	33	15	−0.029	−0.33	15	17	1	0.714→0.701	−0.004
	factor	SCALAR_H	33	15	+0.097	+1.28	18	15	0	0.714→0.683	−0.007
CIFAR-10	dataflow	SCALAR_H	15	30	−0.095	−0.43	8	7	0	0.530→0.758	−0.213
	dataflow	SCALAR_H	15	20	+0.043	+0.28	10	5	0	0.613→0.499	−0.143
	factor	SCALAR_H	15	20	+0.073	+0.39	7	8	0	0.613→0.651	−0.465
Two Moons	dataflow	KL_TRAJ	100	50	−0.008	−0.46	16	21	63	0.703→0.704	+0.043
	factor	KL_TRAJ	100	50	−0.016	−0.77	18	29	53	0.703→0.688	+0.098
	dataflow	SCALAR_H	300	50	+0.003	+0.35	65	66	169	0.704→0.718	−0.189
	dataflow	SCALAR_H	100	50	+0.016	+0.92	25	22	53	0.703→0.702	−0.188
	factor	SCALAR_H	100	50	+0.026	+1.43	26	20	54	0.703→0.681	−0.156
Spirals	dataflow	KL_TRAJ	100	50	+0.294***	+3.79	39	19	42	2.333→2.179	+0.085
	factor	KL_TRAJ	100	50	+0.066	+0.90	29	33	38	2.333→2.399	+0.141
	dataflow	SCALAR_H	300	50	+0.433***	+5.39	136	57	107	2.452→1.928	−0.310
	dataflow	SCALAR_H	100	50	+0.412***	+3.45	43	21	36	2.333→1.800	−0.301
	factor	SCALAR_H	100	50	+0.580**	+3.24	52	20	28	2.333→1.691	−0.263
Circles	dataflow	KL_TRAJ	100	50	−0.016	−0.71	25	22	53	1.223→1.219	+0.026
	factor	KL_TRAJ	100	50	−0.046 $\cdot$	−1.88	25	33	42	1.223→1.214	+0.048
	dataflow	SCALAR_H	100	50	−0.060*	−2.30	22	32	46	1.223→1.230	−0.213
	factor	SCALAR_H	100	50	−0.030	−1.14	29	30	41	1.223→1.229	−0.152